# AI-Based Cattle Feed and Feeding Monitoring: A Comprehensive Report on Deep Learning, Sensing and Edge Technologies (2026 Literature)

*A synthesis of ten recent studies, with comparative analysis, research gaps, Sustainable Development Goal mapping and a roadmap for future work*

## Abstract

Feed is the single largest operating cost in beef and dairy production, and the way cattle consume it shapes growth, milk yield, health, welfare and the environmental footprint of every kilogram of meat or milk produced. Traditional ways of monitoring feeding, such as periodic weighing of feed offered and refused, visual observation by stockpersons, and herd-level averages, are labour intensive, coarse and rarely able to describe what an individual animal actually eats. Over the last few years, advances in deep learning, low-cost computing, infrared and RGB imaging, wearable accelerometers and edge hardware have created new ways to observe feeding automatically.

This report reviews ten studies published in 2026 that apply artificial intelligence (AI) to cattle feed and feeding monitoring. The studies span machine-learning prediction of feed intake from electronic-feeder data, camera-based detection of feeding behaviour and animal identity, forage density detection with lightweight YOLO models, behaviour recognition in night-time video, accelerometer-based classification with random forests, edge-compressed neural networks on wearable devices, and one broad review of smart feedlot technologies. Each study is summarised in terms of method, advantages and limitations, and the studies are then compared along common dimensions.

Three main findings emerge. First, most systems recognise *that* an animal is feeding but not *how much* it ate; only a small number connect behaviour to feed weight, intake or forage availability. Second, individual-level, real-time, low-cost intake estimation without per-animal hardware remains unsolved. Third, validation is weak: datasets are small and drawn from single sites, and accuracy can collapse when models meet unseen animals or farms. The report maps the literature to five United Nations Sustainable Development Goals (2, 9, 12, 13 and 15), discusses practical implications for farms of different sizes, and proposes a research roadmap centred on multimodal fusion, shared benchmarks and field-scale validation.

**Keywords:** precision livestock farming; feed intake; feeding behaviour; deep learning; YOLO; computer vision; accelerometer; edge AI; feedlot; dairy; beef cattle

# 1. Introduction

## 1.1 Why feed monitoring matters

Feed typically represents the largest share of the cost of producing beef and milk. Small improvements in feed efficiency therefore translate into large economic gains, while waste, over-feeding and under-feeding carry direct financial and environmental penalties. Feed intake is also one of the most sensitive indicators of an animal's state. A drop in intake or a change in the timing of meals is often among the earliest signs of disease, heat stress, lameness, social stress or poor ration quality, frequently appearing before clinical symptoms are visible to a human observer.

Feeding monitoring has three interlocking goals:

- **Quantity:** how much feed was offered, consumed and left over, at the level of the pen and, ideally, the individual animal.
- **Behaviour:** when, how often, for how long and in what pattern animals feed, including competition at the bunk.
- **Availability and quality:** how much forage is present on pasture or in the bunk and in what condition.

Achieving all three at scale, continuously and at low cost is the central promise of AI-enabled precision livestock farming (PLF).

## 1.2 From manual observation to AI

For most of the history of livestock production, feeding was monitored by people. Stockpersons judged bunk scores, weighed refusals and walked pens. Research herds used specialised equipment such as electronic feeders with radio-frequency identification (RFID), but these systems are expensive and rarely found on commercial farms. The arrival of inexpensive cameras, single-board computers such as the Raspberry Pi, compact object detectors such as the YOLO family, and wearable microcontrollers has changed what is possible. Models can now locate heads, classify posture, recognise individual animals and detect behaviours in video, or classify behaviour from accelerometer streams on a collar or ear tag.

## 1.3 Aim and objectives of this report

The aim of this report is to provide a structured, critical synthesis of recent AI-based work on cattle feed and feeding monitoring. Specifically, the report seeks to:

1. Describe the technological approaches currently used, and what each can and cannot measure.

2. Summarise ten recent (2026) studies in terms of methodology, advantages and limitations.
3. Compare the studies across sensing modality, learning method, output type, deployment target and validation strength.
4. Identify research gaps and practical barriers to adoption.
5. Map the literature to the United Nations Sustainable Development Goals.
6. Propose directions for future research and development.

### 1.4 Scope and method of the review

The report is based on ten studies compiled from article pages and abstracts and recorded in a structured literature table. Each entry lists authors, year, journal, month, methodology and technology, advantages and limitations. The limitations are the compiler's own assessment drawn from those sources rather than statements made by the original authors. Some details, such as exact publication months for certain items, the journal title for one preprint-style entry, and certain accuracy figures, could not be confirmed and are flagged where they matter. Readers should verify all figures against the full text before citing them.

This is a narrative, qualitative synthesis, not a systematic review with a formal search protocol. The sample of ten studies is small and was selected for relevance to feed and feeding monitoring; conclusions about the field as a whole should be treated as indicative rather than definitive.

### 1.5 Structure of the report

Section 2 gives background on feeding monitoring concepts and traditional methods. Section 3 describes the main technology families. Section 4 reviews each of the ten studies. Section 5 compares them. Section 6 sets out research gaps. Section 7 maps the work to the Sustainable Development Goals. Section 8 discusses practical implications. Section 9 proposes future directions. Section 10 notes the limitations of this review, and Section 11 concludes.

## 2. Background: Feed, Feeding Behaviour and Their Measurement

### 2.1 Key concepts

**Feed intake.** The mass of feed (usually expressed as dry matter) consumed by an animal over a period, typically a day. It is the product of how often an animal eats, how long each meal lasts and how fast it eats.

**Feeding behaviour.** The temporal pattern of feeding: number of visits to the bunk or feeder, meal duration, intervals between meals, time of day and bout structure. Behaviour is a proxy for intake but not identical to it.

**Feed efficiency.** The relationship between feed consumed and product obtained, such as weight gain or milk. Measuring it at the level of the individual requires individual intake.

**Feed waste.** Feed that is offered but not eaten, spoiled, spilled or trampled. Waste reduction is both an economic and an environmental objective.

**Forage availability.** On pasture or in a bunk, the amount and density of feed present. It governs how much animals can eat and how grazing or feed delivery should be managed.

## 2.2 Why individual-level data are valuable

Pen averages hide large differences between animals. Within a group, some animals eat much more than others, some are displaced from the bunk by dominant animals, and some begin to eat less when they become ill. Individual data allow early disease detection, selection for feed-efficient animals, targeted nutrition and fairer assessment of competition. The cost is that each animal must be identified and its intake measured, which is hard to do with cameras alone and expensive with dedicated feeders.

## 2.3 Traditional monitoring approaches

| Approach | What it measures | Strengths | Weaknesses |
|---|---|---|---|
| Manual observation and bunk scoring | Behaviour and leftover feed by eye | No equipment; flexible | Labour intensive; subjective; not continuous |
| Weighing feed offered and refused | Pen-level intake | Simple, widely understood | Coarse; time delay; no individual data |
| Electronic feeders with RFID and load cells | Individual visit and intake | Accurate and individual | High capital cost; restricts feeding setup; needs maintenance |
| Wearable sensors (accelerometers, collars, ear tags) | Behaviour classes such as eating and rumination | Continuous, individual | Device on every animal; battery and loss issues; no direct quantity |

| Approach | What it measures | Strengths | Weaknesses |
|---|---|---|---|
| Cameras with computer vision | Behaviour, head position, identity, sometimes forage level | Non-invasive; scalable per pen | Occlusion, lighting, dust; quantity hard to infer |

No single method satisfies all goals, which is why the literature increasingly looks towards combinations of sensing and modelling.

### 2.4 The role of AI

AI contributes in three ways. First, it **interprets raw signals**: converting video frames or accelerometer traces into behaviour labels, animal identities or object detections. Second, it **predicts** quantities that are hard to measure directly, such as daily feed intake from electronic-feeder history and weather. Third, it **compresses and deploys** models so that analysis can happen on farm hardware near the sensor, reducing latency, bandwidth and power use. The ten reviewed studies map neatly onto these three roles.

## 3. Technology Families for AI-Based Feed Monitoring

### 3.1 Electronic feeding systems and data-driven intake prediction

Electronic feeders, such as the GrowSafe system used in Ref 1, record each animal's visits and the mass of feed removed. They are considered a reference-quality source of individual intake. Their main drawbacks are cost, the need for dedicated infrastructure and the fact that many commercial farms do not have them.

Because these systems generate large, structured datasets, they are well suited to classical machine learning. Ref 1 trained models on eleven years of data from a feedlot in Idaho together with weather-station data and introduced two environmental indices. Gradient-boosted trees (XGBoost) gave the best predictions of daily intake at animal and pen level. The practical value is not only predicting intake but helping managers plan feed delivery and reduce waste. The limitation is dependence on the very feeders many farms lack, and data from a single facility.

### 3.2 Computer vision with object detection

Camera-based monitoring relies on detectors that find objects of interest in each frame. The YOLO family is popular because it balances speed and accuracy and can be made small enough for embedded hardware. In the reviewed set:

- Ref 3 introduced BFDet-YOLO, a lightweight YOLOv10 variant that jointly detects cow head and eating state and forage density level.
- Ref 5 compared YOLOv8 and YOLOv9 for fine-grained cattle activity classification.
- Ref 10 presented BoviFusionNet, a lightweight detector aimed at edge deployment, with box-balanced learning and multi-scale fusion.

Object detection gives where and what, frame by frame. It can indicate that a cow's head is at the feed bunk, but it does not weigh the feed.

### 3.3 Pose estimation and fine-grained behaviour

Keypoint or pose models detect body landmarks, enabling inference of posture and activity. Ref 6 used YOLOv8-pose to track calves, localise them and recognise feeding and drinking. Pose data provide richer information than boxes (for example, head lowered towards a feeder), but multiple similar positions along a fence can be hard to separate, and occlusion by other animals remains a challenge.

### 3.4 Infrared and low-light imaging

Cattle feed around the clock, so systems that fail at night miss a large share of behaviour. Two reviewed studies address darkness: Ref 2 uses motion-triggered infrared cameras on a low-cost Raspberry Pi, and Ref 7 designs a network (STED-Net) with a spatio-temporal enhancement module for dark video. Infrared capture is hardware-based; enhancement networks are software-based. Each has trade-offs in cost, image quality and computing demand.

### 3.5 Animal identification

Individual intake needs individual identity. RFID tags are the standard approach on electronic feeders. Ref 2 explores image-based identity classification, using a VGG16 model to classify both head position and animal ID, and matches feeding events to the weight of the feed bin to estimate intake per event. Image-based identification is attractive because it avoids extra tags, but it is sensitive to training data diversity, viewpoint and changes in appearance over time.

### 3.6 Wearable sensors and classical machine learning

Accelerometers on the nose, ear or neck produce characteristic signals for eating, rumination and movement. Ref 8 applied Random Forest classifiers to features from five datasets, aiming for a standard pipeline. This is machine learning, not deep learning, but it is computationally light and suitable for battery-powered devices. Sensor-based approaches offer continuous, individual data without line-of-sight limitations, but each animal must carry a device.

## 3.7 Edge AI and model compression

Sending raw video or sensor data to the cloud is costly and sometimes impossible in rural areas. Edge AI runs models on or near the device. Ref 9 compressed a network using a graph neural network approach so it could run on a wearable low-power microcontroller, reducing latency and power use. Refs 3 and 10 take a similar philosophy for vision, building very small detectors. The central tension is accuracy versus efficiency: Ref 7 achieves good night-time results but at about 292 ms per sample, too slow for strict real-time edge use.

### 3.7.1 Why edge matters for farms

Farms often have limited connectivity, harsh environments and tight budgets. On-device processing reduces data transfer, keeps systems working during network outages, supports faster alerts and can improve privacy by keeping raw video local. However, long-term field results for lightweight models are largely absent from the reviewed studies.

## 3.8 Reviews and system-level perspectives

Ref 4 provides a broad narrative review of smart feedlot technologies, covering RFID, automated weighing, machine vision, smart troughs, feed and water intake monitoring and AI decision support, drawing on around 350 publications of which 117 were cited. It argues for priorities such as edge-cloud architecture and multi-sensor fusion, and highlights cost, durability and data governance as continuing barriers.

## 3.9 Summary of technology families

| Technology | Typical output | Measures feed quantity? | Needs device per animal? | Typical deployment |
|---|---|---|---|---|
| Electronic feeder plus ML (Ref 1) | Daily intake per animal and pen | Yes (via feeder data) | Tag per animal and feeder | Research and large feedlots |
| IR camera plus CNN plus bin weight (Ref 2) | Head position, ID, intake per event | Yes (via bin weight) | No | Farm bunk, offline analysis |
| Lightweight YOLO for forage and behaviour (Ref 3) | Eating behaviour and forage density level | Level only, not kg | No | Pasture, edge capable |

| Technology | Typical output | Measures feed quantity? | Needs device per animal? | Typical deployment |
|---|---|---|---|---|
| YOLO activity detection (Ref 5) | Activity classes | No | No | Farm cameras |
| Pose estimation (Ref 6) | Posture, feeding and drinking | No | No | Experimental pens |
| Night video network (Ref 7) | Six behaviours including feeding | No | No | Night barn video, not edge ready |
| Accelerometer Random Forest (Ref 8) | Eating, rumination, movement | No | Yes | Wearables |
| GNN-compressed network (Ref 9) | Behaviour on device | Not confirmed | Yes | Wearable microcontroller |
| Lightweight detector (Ref 10) | Eating, standing, lying | No | No | Edge, barn and pasture |

# 4. Review of the Ten Studies

This section summarises each study using a common template: context and objective, methodology, strengths, limitations and relevance to feed monitoring. Figures are those recorded in the source literature table; items marked as not confirmed or not retrieved could not be verified.

## 4.1 Ref 1: AI-Based Framework to Predict Animal and Pen Feed Intake in Feedlot Beef Cattle

**Citation:** Maia, Hall, Milan and Teixeira (2026), *Smart Agricultural Technology*. Month of publication not confirmed.

**Context and objective.** Feedlot managers need to know how much each animal and pen will eat in order to plan feed delivery, control costs and limit waste. Intake varies with weather, animal characteristics and day-to-day management, which makes it hard to predict from simple rules.

**Methodology.** The study applied machine learning to eleven years of electronic-feeder (GrowSafe) data from an Idaho feedlot, combined with weather-station data. Two new environmental indices were introduced, the InComfort-Index and the EASI-Index, to

represent how climate affects cattle. Several models were tested and XGBoost gave the best predictions of daily intake per animal and per pen.

**Strengths.** The framework predicts intake at both animal and pen level, with a reported RMSE of 1.38 kg per day per animal and 0.14 kg per day-animal at pen level. It explicitly accounts for climate, and it supports feed-waste reduction and feed delivery planning. The very long data record is unusual and gives the models a rich picture of seasonal variation.

**Limitations.** The data come from a single feedlot, so generalisation to other regions, breeds and management systems is untested. The approach depends on electronic feeding systems that many farms lack. The InComfort-Index predicted feed intake poorly, and the EASI-Index needs feeding-behaviour data, which creates a circular dependency for farms without that data.

**Relevance.** Ref 1 is the clearest example in the set of AI that estimates *quantity*, but it does so by modelling, not by observing animals directly. It shows what good intake prediction looks like when instrumentation is available and sets a benchmark for lower-cost approaches.

## 4.2 Ref 2: Automated 24-h Cattle Feeding Behaviour Analysis Using Raspberry Pi, Infrared Imaging and Deep Learning

**Citation:** Wager-Jones, Butler, Harris, Rutter, Bleach and Behrendt (2026), *Biosystems Engineering*. Month not confirmed.

**Context and objective.** The aim was a low-cost, non-invasive system that monitors feeding around the clock and links behaviour to individual animals and to feed weight.

**Methodology.** A Raspberry Pi with motion-triggered infrared cameras was installed at the feed bunk. VGG16 models, implemented with TensorFlow and Keras, classified head position and animal identity. Muzzle-to-feed events were matched to feed bin weight to estimate the amount eaten per event.

**Strengths.** The system works day and night under real farm conditions, offers a non-invasive alternative to wearables, and connects behaviour to individual animals and feed weight. Using inexpensive hardware makes it a realistic template for commercial adoption.

**Limitations.** Training and inference were run offline, not in real time. The approach needs large and varied training data and depends on bin weight data and careful camera placement.

**Relevance.** Ref 2 is one of only three studies that link behaviour to feed quantity, and it is the only camera-based system in the set that attempts per-event intake for identified animals. It is therefore the closest match to the unmet need described in the research gap

analysis, although it is not yet real time.

## 4.3 Ref 3: Multi-Object Detection of Forage Density and Dairy Cow Feeding Behaviour (BFDet-YOLO)

**Citation:** Liu et al. (2026), *Sensors*, 26(4), 1273. February 2026.

**Context and objective.** Pasture-based systems need to know both whether cows are eating and how much forage is available. Most systems address one or the other.

**Methodology.** BFDet-YOLO is a lightweight variant of YOLOv10 with feature fusion, an occlusion-handling module (SEAM) and an improved DIoU box loss. It jointly detects cow head and eating state and forage density level on a custom pasture dataset.

**Strengths.** The model links feed availability with feeding behaviour in a single network. It is very small, with 1.85 million parameters and a 4.45 MB size, making it suited to edge devices. Reported performance is precision 94.2%, recall 93.5% and mAP@50 of 95.7%.

**Limitations.** Forage is graded by density level, not weighed in kilograms. The mAP@50-95 is lower at 70.7%, indicating that localisation is less precise under stricter overlap thresholds. The dataset is pasture specific, with limited multi-farm validation.

**Relevance.** Ref 3 shows how availability and behaviour can be combined in one lightweight model. It moves toward intake understanding without weighing feed, and its small size makes field deployment plausible.

## 4.4 Ref 4: Modernizing Livestock Operations: Smart Feedlot Technologies and Their Impact

**Citation:** Dao, Gostar, Tennakoon, Chuah and Bab-Hadiashar (2026), *Animals*, 16(8), 1244. April 2026.

**Context and objective.** The article reviews the technologies that make up a smart feedlot and their effect on productivity, welfare and sustainability.

**Methodology.** A narrative review of about 350 publications, of which 117 are cited, covering RFID, automated weighing, machine vision, smart troughs, feed and water intake monitoring and AI decision support.

**Strengths.** It covers the whole feedlot pipeline, links precision feeding to better feed use and less waste, and sets out priorities such as edge-cloud design and multi-sensor fusion. It is a useful map of the field and a source of context for the more specialised experimental papers.

**Limitations.** As a review it reports no new experiments. Intake-monitoring systems are mostly validated in trials rather than in commercial multi-breed herds. Cost, durability and data governance remain barriers.

**Relevance.** Ref 4 is the main source for adoption barriers and thin evidence on cost and waste savings, which is why it features in the research gap on adoption.

### 4.5 Ref 5: Cattle Farming Activity Monitoring Using an Advanced Deep Learning Approach

**Citation:** Asim et al. (2026), *Sensors*, 26(3), 785. January 2026.

**Context and objective.** To classify fine-grained cattle activities from farm camera images using modern detectors.

**Methodology.** A comparison of YOLOv8 and YOLOv9 detectors.

**Strengths.** YOLOv8-L reached 91.11% mAP versus 90.23% for YOLOv9-E. Inference time was stable, and the approach is practical for camera-based monitoring.

**Limitations.** The study classifies behaviour only and does not estimate feed quantity. The gap between YOLO variants is small, and generalisation across farms and breeds was not shown.

**Relevance.** Ref 5 represents the common pattern in the field: strong detection performance on behaviour, with no direct link to intake.

### 4.6 Ref 6: Integrating YOLOv8-pose for Localisation and Behaviour Tracking of Calves

**Citation:** Jahn, Düpjan, Röttgen, Jordt, Albrecht and Homeier-Bachmann (2026), *Frontiers in Animal Science*. January 2026.

**Context and objective.** To track calf location and behaviour in precision livestock farming using pose estimation.

**Methodology.** A YOLOv8-pose keypoint model tracks calf posture and position and recognises feeding and drinking from video.

**Strengths.** Video analysis is fast. Feeding periods matched manual annotation closely, with about 99% overlap, and the system also provides posture data.

**Limitations.** The study took place in an experimental setting. It could not distinguish two feeding positions at the same fence, and it addressed calves only.

**Relevance.** Ref 6 shows the value of pose data for feeding and drinking detection and highlights the difficulty of fine spatial distinctions at crowded feeding fences.

## 4.7 Ref 7: Beef Cattle Behaviour Recognition Based on Nighttime Farm Videos (STED-Net)

**Citation:** Han, Zhang, Zhang, Cao, Sun, Jia, Wu, Huang and Zhang (2026), *Animals*, 16(12), 1881. June 2026.

**Context and objective.** To recognise behaviour in dark farm video, where standard models struggle.

**Methodology.** STED-Net combines a spatio-temporal dark-video enhancement module with a dynamic fusion block. A new dataset of 1,097 night clips from 30 Angus cattle covers six behaviours including feeding.

**Strengths.** The model handles dark video, with precision 88.47%, recall 80.18%, accuracy 83.80% and F1 of 84.12%. Feeding was among the best classes, with an F1 of 87.0%.

**Limitations.** Processing takes about 292 ms per sample, too slow for edge or strict real-time use. Classes are imbalanced, data come from a single farm and no feed quantity is estimated.

**Relevance.** Ref 7 fills the night-time gap in software, in contrast with the hardware approach of Ref 2, and quantifies the accuracy-versus-speed trade-off.

## 4.8 Ref 8: Advancing Standardised Cattle Behaviour Classification with a Random Forest Model

**Citation:** James, Rial, Giordano and Erickson (2026), *Frontiers in Animal Science*. January 2026.

**Context and objective.** To build a standard classification pipeline for accelerometer data that works across datasets.

**Methodology.** A Random Forest classifier on accelerometer features from nose, ear and neck sensors, evaluated across five datasets, classifying eating, rumination and movement.

**Strengths.** One standard pipeline works across datasets with an average accuracy of 92.7% under an 80/20 split. Compute demand is low, suiting battery-powered devices.

**Limitations.** It is machine learning rather than deep learning. Accuracy drops under stricter validation, for example from 93.4% to 61.5% under leave-one-individual-out testing. Transfer between datasets is weak and each animal needs a sensor.

**Relevance.** Ref 8 is the strongest cautionary result in the set. It shows that headline accuracy can overstate real-world performance when models meet unseen animals.

## 4.9 Ref 9: Cattle Behaviour Recognition Based on Graph Neural Network Compression on the Edge

**Citation:** Liu, Song, Xin, Rong, Gao, Wang and Zhang (2026), *Animals*, 16(3), 430. January 2026.

**Context and objective.** To run behaviour recognition on a wearable low-power microcontroller for large free-grazing herds.

**Methodology.** An S-ResNet takes single-frame sensor input and is compressed with a graph neural network, then run on-device.

**Strengths.** On-device inference lowers latency and power use and extends battery life, which suits large herds in remote areas.

**Limitations.** A device is needed on every animal. Per-behaviour results and whether feeding is classified separately could not be retrieved.

**Relevance.** Ref 9 demonstrates the feasibility of on-device deep learning for wearables, though its direct value for feeding monitoring is uncertain until feeding performance is confirmed.

## 4.10 Ref 10: BoviFusionNet: A Lightweight Edge-Deployable AI System for Cattle Behaviour Recognition

**Citation:** Li, Zhang, Ren, He, Wang, Lv, Lin, Du and Chen (2026). Open-access journal, title not confirmed. July 2026.

**Context and objective.** To recognise behaviour in dense, occluded cattle scenes on edge hardware.

**Methodology.** A lightweight object detector with box-balanced learning, improved downsampling, multi-scale fusion and fine-grained discrimination, trained on a box-balanced dataset for eating, standing and lying.

**Strengths.** It is designed for edge deployment on barns and pastures and targets dense, occluded cattle and visually similar behaviours.

**Limitations.** Only three behaviours are covered, accuracy figures were not retrieved, and it detects eating but not feed quantity.

**Relevance.** Ref 10 reflects the trend towards compact, deployable detectors, but its performance claims should be checked against the full text.

# 5. Comparative Analysis

## 5.1 Study characteristics at a glance

| Ref | Primary input | Method | Main output | Feed quantity link | Edge or real-time | Night capable |
|---|---|---|---|---|---|---|
| 1 | Electronic-feeder and weather data | XGBoost and other ML | Daily intake per animal and pen | Yes (prediction) | Not applicable | Not applicable |
| 2 | Infrared camera and bin weight | VGG16 CNN | Head position, ID, intake per event | Yes (bin weight) | Offline | Yes |
| 3 | Pasture camera images | BFDet-YOLO (YOLOv10) | Eating and forage density | Density level | Edge suited | Not stated |
| 4 | Literature | Narrative review | Technology overview | Discussed | Discussed | Discussed |
| 5 | Farm camera images | YOLOv8, YOLOv9 | Activity classes | No | Stable inference | Not stated |
| 6 | Calf video | YOLOv8-pose | Posture, feeding, drinking | No | Fast video analysis | Not stated |
| 7 | Night video | STED-Net | Six behaviours | No | No (292 ms per sample) | Yes |
| 8 | Accelerometers | Random Forest | Eating, rumination, movement | No | Low compute | Yes (sensor) |
| 9 | Wearable sensor | GNN-compressed S-ResNet | Behaviour on device | Not confirmed | Yes | Yes (sensor) |
| 10 | Barn and pasture images | Lightweight detector | Eating, standing, lying | No | Edge designed | Not stated |

## 5.2 Reported performance metrics

The studies use different metrics, which complicates direct comparison.

| Ref | Metric type | Reported values |
|---|---|---|
| 1 | Regression error | RMSE 1.38 kg/day per animal; 0.14 kg/day-animal at pen level |
| 3 | Detection | Precision 94.2%, recall 93.5%, mAP@50 95.7%, mAP@50-95 70.7% |

| Ref | Metric type | Reported values |
|---|---|---|
| 5 | Detection | YOLOv8-L 91.11% mAP; YOLOv9-E 90.23% mAP |
| 6 | Temporal agreement | About 99% overlap with manual annotation |
| 7 | Classification | Precision 88.47%, recall 80.18%, accuracy 83.80%, F1 84.12%; feeding F1 87.0% |
| 8 | Classification | Average accuracy 92.7%; 93.4% falling to 61.5% under leave-one-individual-out |
| 2, 4, 9, 10 | Not available or not retrieved | Figures not reported in the source table |

Because regression error, detection precision and classification accuracy measure different things, no ranking of the studies is possible from these figures. Within comparable categories, Ref 3 reports the strongest detection scores, but the dataset is pasture specific and tested on limited sites.

## 5.3 Sensing modality: camera versus wearable

Vision systems (Refs 2, 3, 5, 6, 7, 10) are non-invasive and scale by pen rather than by animal, but struggle with occlusion, lighting and identity. Sensor-based systems (Refs 8, 9) provide continuous individual data and are unaffected by line of sight, but need a device on each animal, with cost, battery and maintenance implications. Ref 1 relies on dedicated electronic feeders, which provide the best quantitative data but the least flexibility.

## 5.4 Behaviour versus quantity

A clear dividing line runs through the set. Refs 1 to 3 connect their outputs to feed quantity, forage density or intake. Refs 5 to 10 recognise feeding or eating as one behaviour class among several, without estimating how much was eaten. Because behaviour duration is only a weak proxy for intake (animals can eat quickly or slowly, and bunk conditions vary), behaviour-only systems cannot replace intake measurement.

## 5.5 Accuracy versus efficiency

The more elaborate the model, the harder it is to run on a farm. Ref 7 is accurate on dark video but slow. Refs 3, 9 and 10 aim for small models that run on edge hardware, but they lack long-term field deployment results. The practical question is not which model scores highest in the laboratory but which meets an acceptable accuracy at an affordable cost and latency over months of real use.

## 5.6 Validation strength

Validation is generally weak. Single-farm datasets (Refs 1, 3, 7), experimental pens (Ref 6) and random train-test splits that allow information from the same animal to appear in both training and test sets can all inflate apparent performance. Ref 8 is a rare case in which stricter validation was applied, and the drop in accuracy from 93.4% to 61.5% shows why it matters. Readers should treat headline accuracy figures in this field with caution unless the validation design separates individuals, farms and time periods.

## 5.7 Strengths and weaknesses by approach

| Approach | Main strengths | Main weaknesses |
|---|---|---|
| Feeder data plus ML | Highest quantitative accuracy; climate aware | Needs electronic feeders; single-site |
| Infrared camera plus ID plus bin weight | Day and night; individual; low cost | Offline; data hungry; placement dependent |
| Lightweight YOLO detectors | Small, fast, edge suited | Behaviour or density only; limited cross-farm testing |
| Pose estimation | Rich posture data | Fine position ambiguity; experimental |
| Night-video networks | Handle darkness | Slow; single farm |
| Accelerometers plus ML | Individual, continuous, light compute | Device per animal; poor transfer |
| Compressed networks on wearables | Low power, low latency | Device per animal; feeding results unclear |

## 5.8 Cross-cutting themes

**Theme 1: The lightweight-model trend.** Three of the ten studies (Refs 3, 9 and 10) are explicitly about compact models. This reflects a recognition that farm deployment is constrained by hardware, power and connectivity. The trend is encouraging, but compactness has been demonstrated mainly through parameter counts and model size, not through months of continuous operation in dust, moisture and temperature extremes.

**Theme 2: Dark and difficult conditions.** Only two studies (Refs 2 and 7) deal with night-time monitoring, yet cattle feed throughout the day and night, and night-time behaviour may be where welfare problems first appear. Dust, rain, glare and occlusion at bunks remain largely unaddressed.

**Theme 3: From behaviour to decision.** Few studies connect their outputs to a management decision such as adjusting a ration, raising a health alert or scheduling delivery. Ref 1 comes closest by supporting delivery planning. A system that merely labels behaviour has limited value unless it is wired into decisions.

**Theme 4: Data scarcity and diversity.** Labelled datasets are expensive to create. Studies draw on single farms, a few dozen animals or experimental pens. Model performance on a new farm, breed, coat colour or housing system is rarely tested.

# 6. Research Gaps

Synthesising the limitations reported across the ten reviewed studies, nine gaps emerge. Each is described below with the evidence behind it and a suggested response.

## 6.1 Behaviour is detected, but feed quantity is not measured

Most studies (Refs 5 to 10) recognise feeding or eating as a behaviour but do not estimate how much feed was eaten or left. Only Refs 1 to 3 link behaviour to feed weight, intake or forage density. Since intake is the quantity that matters for efficiency, health and waste, this is the most important gap.

*Suggested response:* combine behaviour recognition with bunk-level weight or volume sensing, for example load cells under feed bins, depth cameras that estimate feed height, or image-based estimation of feed remaining, and validate each against weighed refusals.

## 6.2 Individual-level intake is hard to obtain

Ref 1 predicts intake well but relies on electronic feeders. Ref 2 links feeding events to bin weight but works offline. Low-cost, real-time, per-animal intake from cameras is still unsolved.

*Suggested response:* develop pipelines that integrate identity recognition, head-at-feeder detection and bunk weight change in real time on edge hardware, and test them in competitive group-feeding settings where several animals share a bunk.

## 6.3 Limited multimodal fusion

Vision systems (Refs 2, 3, 5, 6, 7, 10) and sensor or wearable systems (Refs 1, 8, 9) are mostly studied separately. Few combine camera, weight and accelerometer data. Each modality fails differently: cameras suffer from occlusion and darkness, accelerometers from poor transfer between animals, and weight sensors from inability to say which animal ate. Fusion could cancel out these weaknesses.

*Suggested response:* design fusion architectures that use accelerometer data to confirm cameras during occlusion, and cameras to resolve identity when sensors are missing, with explicit handling of missing modalities.

### 6.4 Small, single-site datasets

Studies use one feedlot (Ref 1), one pasture dataset (Ref 3), 30 Angus cattle (Ref 7) or experimental calf pens (Ref 6). This limits generalisation across breeds, lighting and farm layouts.

*Suggested response:* build multi-farm, multi-breed, multi-season datasets with standard annotation, and report performance by farm and breed.

### 6.5 Weak validation across animals and datasets

Ref 8 shows accuracy falling from 93.4% to 61.5% when tested on unseen individuals, and transfer between datasets was often poor. Without individual-level and farm-level hold-out testing, reported accuracy is likely to be optimistic.

*Suggested response:* adopt leave-one-animal-out and leave-one-farm-out evaluation as standard, and report both alongside random-split results.

### 6.6 Day, night and outdoor robustness

Only Refs 2 and 7 address night-time monitoring. Dust, rain and occlusion at feed bunks and pastures remain hard.

*Suggested response:* collect data across weather and lighting conditions, use sensor housing designed for farm environments, and test models under deliberate degradation such as lens dirt or heavy occlusion.

### 6.7 Accuracy versus edge efficiency

Ref 7 takes about 292 ms per sample, too slow for edge use, while lightweight models (Refs 3, 9, 10) lack long-term field deployment results.

*Suggested response:* report latency, power draw and memory on named hardware, use distillation and quantisation to shrink accurate models, and run multi-month field trials.

### 6.8 Wearable burden

Sensor approaches (Refs 8, 9) need a device on every animal, raising cost and maintenance for large herds. Batteries, tag loss, animal welfare and device attachment all add to the burden.

*Suggested response:* prefer sensing that scales by pen where possible, and reserve wearables for cases where their continuous individual data justify the cost.

### 6.9 No unified benchmark and thin adoption evidence

Datasets, behaviour classes and metrics (mAP, accuracy, F1, RMSE) differ between studies, and few report cost, feed-waste savings or usability for smaller farms (Ref 4). Without common benchmarks, progress cannot be tracked, and without cost-benefit evidence, farmers cannot decide whether to invest.

*Suggested response:* create open benchmark datasets and a shared set of behaviour definitions and metrics, and require economic and usability reporting in field studies.

### 6.10 Gap summary table

| Gap | Evidence in this review | Priority |
|---|---|---|
| Quantity not measured | Refs 5 to 10 detect behaviour only | High |
| Individual intake at low cost, real time | Refs 1 and 2 each fall short | High |
| Multimodal fusion | Vision and sensor work studied separately | High |
| Small single-site data | Refs 1, 3, 6, 7 | High |
| Weak cross-animal validation | Ref 8 (93.4% to 61.5%) | High |
| Day, night, outdoor robustness | Only Refs 2 and 7 address night | Medium |
| Accuracy versus edge efficiency | Ref 7 (292 ms); Refs 3, 9, 10 lack field trials | Medium |
| Wearable burden | Refs 8, 9 | Medium |
| Benchmarks and adoption evidence | Differing metrics; Ref 4 | High |

The priority column is this report's judgement based on how many studies are affected and how directly the gap blocks practical intake monitoring.

## 7. Mapping to the United Nations Sustainable Development Goals

The reviewed literature aligns with five Sustainable Development Goals (SDGs).

| SDG | Relevant studies | Mapping rationale |
|---|---|---|
| SDG 2: Zero Hunger | Refs 1 to 10 (all) | Better feed and feeding monitoring improves livestock productivity and the efficiency of milk and meat supply |
| SDG 9: Industry, Innovation and Infrastructure | Refs 1 to 4, 7, 9, 10 | Edge AI, lightweight YOLO models, electronic feeding and smart-feedlot technology are innovations in agricultural infrastructure |
| SDG 12: Responsible Consumption and Production | Refs 1 to 4 | Measuring feed intake, leftovers and forage availability helps cut feed waste and supports precise feeding |
| SDG 13: Climate Action | Refs 1, 4 | Precision feeding and climate-adaptive feed management can lower emissions per unit of beef or milk |
| SDG 15: Life on Land | Refs 3, 8 | Pasture feed monitoring and behaviour-based grazing management (for example virtual fencing) support sustainable use of grazing land |

## 7.1 SDG 2: Zero Hunger

All ten studies serve the broad aim of making animal protein production more efficient and resilient. Efficient feed use means more milk and meat per unit of resource, and early detection of reduced intake helps protect animal health and productivity. The link to hunger is indirect, running through the stability and efficiency of food supply chains.

## 7.2 SDG 9: Industry, Innovation and Infrastructure

Electronic feeders, edge-deployed lightweight networks and smart-feedlot systems are examples of new agricultural infrastructure. The reviewed work shows innovation in both hardware (infrared Raspberry Pi systems, wearable microcontrollers) and software (compressed networks, lightweight detectors). A remaining challenge is making this infrastructure affordable to smaller farms.

### 7.3 SDG 12: Responsible Consumption and Production

Knowing how much feed is offered, eaten and left over is the foundation of reducing waste. Refs 1 to 4 contribute through intake prediction, intake-per-event estimation, forage density and system-level planning. Evidence on actual waste reductions is thin and should be a priority for field studies.

### 7.4 SDG 13: Climate Action

Feed efficiency affects emissions intensity, the emissions per kilogram of product. Ref 1 explicitly accounts for climate and weather in intake prediction, and Ref 4 discusses precision feeding in feedlots. The link to emissions is plausible but is not quantified in the reviewed material.

### 7.5 SDG 15: Life on Land

Pasture monitoring (Ref 3) and behaviour-based grazing management (Ref 8) can support more sustainable use of grazing land by informing where and when animals graze, helping avoid overgrazing and supporting land health.

## 8. Practical Implications and Deployment Considerations

### 8.1 Choosing an approach by farm type

The right monitoring strategy depends on the production system, the budget and the question being asked. The reviewed literature suggests the following broad guidance, which reflects this report's interpretation rather than conclusions stated by the original authors.

| Farm type | Main need | Most relevant approaches from the review |
|---|---|---|
| Large feedlot with electronic feeders | Predict intake, plan delivery, cut waste | Ref 1 style ML on feeder and weather data; Ref 4 for system design |
| Commercial dairy or beef barn with bunk feeding | Individual feeding behaviour, 24-hour coverage | Ref 2 style infrared cameras with ID and bin weight; Ref 7 for night video; Ref 10 for edge detection |

| Farm type | Main need | Most relevant approaches from the review |
|---|---|---|
| Pasture-based dairy | Forage availability and grazing behaviour | Ref 3 forage and behaviour detection; Ref 8 and Ref 9 wearables for free-ranging animals |
| Calf rearing | Feeding and drinking tracking | Ref 6 pose-based tracking |
| Small or mixed farm with limited budget | Low-cost alerts for changes in feeding | Low-cost Raspberry Pi systems (Ref 2) and small detectors (Refs 3, 10); wearables only where justified |

## 8.2 Economic considerations

The reviewed studies describe technical performance far more than economic return. Ref 4 notes that cost, durability and data governance remain barriers. For a farm considering adoption, a sensible cost-benefit analysis would include:

- **Capital cost:** cameras, computing hardware, load cells, wearables, mounting and networking.
- **Running cost:** power, connectivity, replacement of damaged or lost devices, software updates and technical support.
- **Labour cost:** time to install, calibrate, label data and respond to alerts.
- **Expected benefit:** reduced feed waste, earlier disease detection, improved feed efficiency, better ration planning and reduced labour for observation.

Because few studies report cost or waste savings, farms must currently estimate benefits themselves. This is a significant evidence gap for smaller operations.

## 8.3 Data quality, labelling and maintenance

AI systems are only as good as their data. Camera-based models need labelled images or clips, and labelling is slow and requires expertise. Sensor-based systems need reliable ground truth, often from video observation. Over time, conditions change: animals grow, coats and lighting change, camera lenses get dirty, and feeding routines shift. Models therefore need monitoring and periodic retraining, a maintenance burden that academic studies rarely discuss.

## 8.4 Animal welfare and ethics

Non-invasive camera systems avoid attaching devices to animals, which is a welfare advantage. Wearables can cause discomfort, injury or stress if poorly fitted and need handling for attachment and removal. In addition, continuous monitoring can reveal welfare problems earlier, but only if alerts are acted on. Systems should be designed to support, not replace, stockperson care and judgement.

## 8.5 Data governance and privacy

Farm data may include video of workers, production data and information of commercial value. Ref 4 flags data governance as a barrier. Farms need clarity on who owns the data, where it is stored, who can access it and how it can be used. Edge processing, which keeps raw video on the farm and transmits only results, can reduce privacy and bandwidth concerns.

## 8.6 Integration with farm management

A monitoring system creates value only when its outputs feed into decisions. Examples include automatically flagging an animal whose feeding visits drop below its own baseline, adjusting the amount of feed delivered to a pen based on predicted intake, or alerting staff when a feed bunk is empty at an unusual time. Interoperability with existing herd-management software, feeding systems and mixers matters as much as model accuracy.

## 8.7 Risks and failure modes

| Risk | Example | Mitigation |
|---|---|---|
| Over-trust in accuracy figures | Ref 8 drop from 93.4% to 61.5% on unseen animals | Insist on individual-level and farm-level validation |
| Behaviour mistaken for intake | Long time at bunk does not mean large intake | Pair behaviour with weight or volume sensing |
| Camera failure | Dust, dirt, condensation, occlusion | Enclosures, cleaning schedules, redundancy |
| Identity errors | Misclassified animal ID | Confidence thresholds, human review of uncertain cases |
| Model drift | Seasonal and growth-related change | Scheduled retraining and monitoring |
| Alert fatigue | Too many false alarms | Tune thresholds, prioritise alerts |

| Risk | Example | Mitigation |
|---|---|---|
| Vendor lock-in | Proprietary formats | Prefer open standards and exportable data |

# 9. Future Directions and Proposed Research Roadmap

## 9.1 Toward real-time, low-cost, individual intake estimation

The central technical challenge is to estimate individual intake in real time at low cost. A plausible architecture, building on Refs 2 and 3, would combine: (a) an edge camera to detect which animal is at the bunk and whether its head is in the feeding zone; (b) a load cell or depth sensor to track bunk feed mass or height; and (c) a time-synchronised model that attributes weight change to the identified animal. Such a system would need to handle several animals feeding at once, which is the hardest part.

## 9.2 Multimodal fusion

Fusing video, weight and accelerometer data could provide redundancy. Where cameras are occluded, an ear-tag accelerometer can confirm eating. Where animals are not tagged, a camera can supply identity. Research should explore architectures that handle missing or noisy inputs gracefully and that are validated on the failure cases, not just on clean data.

## 9.3 Quantifying forage and feed from images

Ref 3 grades forage density but does not weigh it. Future work could calibrate image-based estimates against measured biomass or feed mass, using depth sensing, multispectral imaging or reference objects, so that the output moves from categories to kilograms with known uncertainty.

## 9.4 Open benchmarks and common protocols

The field needs shared datasets with standard annotations for eating, drinking, ruminating and bunk occupancy, recorded across farms, breeds, seasons and lighting. A common evaluation protocol with leave-one-animal-out and leave-one-farm-out splits, together with agreed metrics, would let studies be compared fairly. Reporting latency, power and hardware in a standard format would make accuracy-versus-efficiency trade-offs visible.

## 9.5 Long-term field validation

Lightweight models (Refs 3, 9, 10) need multi-month trials on commercial farms, reporting uptime, maintenance, drift and the effect on management outcomes. Trials should include smaller farms and a variety of housing systems.

## 9.6 Foundation models and self-supervised learning

Labelling cost is a major barrier. Self-supervised pre-training on large amounts of unlabelled farm video, and adapting general-purpose vision models with few labelled examples, could reduce the labelling needed for each new farm. This is a forward-looking suggestion of this report and is not tested in the reviewed studies.

## 9.7 Explainability and farmer trust

Farmers and veterinarians are more likely to act on a system they understand. Outputs should be presented as simple, interpretable indicators, such as an animal's feeding time compared with its own recent baseline, with confidence shown and the underlying video available for review.

## 9.8 Economic and environmental evaluation

Future studies should report the cost of the system, the feed or labour saved, the effect on animal health and productivity, and where possible the change in emissions or waste. Without such data, the sustainability claims mapped in Section 7 remain plausible but unproven.

## 9.9 Roadmap summary

| Horizon | Priority actions |
|---|---|
| Short term (1 to 2 years) | Adopt strict validation; publish multi-farm datasets; report latency and cost; extend night and outdoor testing |
| Medium term (2 to 5 years) | Real-time per-animal intake from camera plus weight fusion; commercial field trials on varied farms; integration with management software |
| Long term (5 years and beyond) | Self-supervised and foundation-model approaches; standard benchmarks and certification; evidence-based economic and emissions assessment |

## 10. Limitations of This Review

This report should be read with the following caveats.

1. **Small and selective sample.** Ten studies from 2026 cannot represent the whole field. Earlier work and other databases were not included.
2. **Abstract-level information.** The underlying literature table was compiled from article pages and abstracts. Limitations are the compiler's assessment and should be verified against the full text before citation.
3. **Unconfirmed details.** Some publication months, one journal title, some performance figures and the per-behaviour results of Refs 9 and 10 could not be confirmed or retrieved.
4. **Non-comparable metrics.** The studies report different metrics on different datasets, so no ranking is attempted.
5. **Interpretive sections.** Practical guidance in Section 8, the priority ratings in Section 6 and the roadmap in Section 9 reflect this report's reasoning and are not findings of the original studies.
6. **No systematic protocol.** The review is narrative and does not follow a formal systematic-review procedure.

## 11. Conclusion

AI-based cattle feed and feeding monitoring is advancing quickly. The ten 2026 studies reviewed here show capable detectors that find cattle heads and eating events, pose models that track calves, networks that work in the dark, classifiers that run on wearable microcontrollers, and machine-learning models that predict feed intake accurately when electronic-feeder data are available. Lightweight architectures show that edge deployment is becoming realistic.

Yet the most valuable output for farmers, how much each animal actually ate, is still largely out of reach for low-cost systems. Most studies recognise eating as a behaviour without weighing feed. Those that do connect behaviour to quantity either depend on expensive electronic feeders or run offline. Datasets are small and often come from a single site, and the one study that tested accuracy on unseen animals saw performance fall sharply. Evidence on cost, waste reduction and usability on smaller farms is thin.

The path forward is clear in outline. Researchers should pursue multimodal fusion that combines identity, behaviour and feed weight in real time on affordable edge hardware; adopt stricter, shared validation standards and open benchmarks; run long field trials across varied farms; and report economic and environmental outcomes. If these steps are taken, AI-based feed monitoring can contribute meaningfully to more efficient, lower-waste and more sustainable livestock production, supporting SDGs 2, 9, 12, 13 and 15.

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